

INFT-1205 – Data Communications & Networking II

Capstone Project

# Port Security Documentation

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## Document Overview

This document serves as brief documentation on layer 2 security configurations on the FL1 and FL2 switches, including port security, DHCP snooping, Dynamic ARP Inspection, and BPDU Guard.

### 1. show port-security – FL1

Used ports have a violation mode of restrict, and unused ports have a violation mode of shutdown.

```
FL1#show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
              (Count)      (Count)      (Count)
-----
Gi1/0/1      2                2            0                Restrict
Gi1/0/3      2                0            0                Restrict
Gi1/0/4      1                0            0                Shutdown
Gi1/0/8      1                0            0                Shutdown
Gi1/0/9      1                0            0                Shutdown
Gi1/0/10     1                0            0                Shutdown
Gi1/0/11     1                0            0                Shutdown
Gi1/0/12     1                0            0                Shutdown
Gi1/0/13     1                0            0                Shutdown
Gi1/0/14     1                0            0                Shutdown
Gi1/0/15     1                0            0                Shutdown
Gi1/0/16     1                0            0                Shutdown
Gi1/0/17     1                0            0                Shutdown
Gi1/0/18     1                0            0                Shutdown
Gi1/0/19     1                0            0                Shutdown
Gi1/0/20     1                0            0                Shutdown
Gi1/0/21     1                0            0                Shutdown
Gi1/0/22     1                0            0                Shutdown
Gi1/0/23     1                0            0                Shutdown
Gi1/0/24     1                0            0                Shutdown
Gi1/0/25     1                0            0                Shutdown
Gi1/0/26     1                0            0                Shutdown
Gi1/0/27     1                0            0                Shutdown
Gi1/0/28     1                0            0                Shutdown
-----
Total Addresses in System (excluding one mac per port)    : 1
Max Addresses limit in System (excluding one mac per port) : 16384
```

## 2. show port-security – FL2

Used ports have a violation mode of restrict, and unused ports have a violation mode of shutdown.

```
FL2#show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
              (Count)      (Count)      (Count)
-----
Gi1/0/1      2                2            0                Restrict
Gi1/0/2      1                0            0                Shutdown
Gi1/0/3      2                0            0                Restrict
Gi1/0/4      1                0            0                Shutdown
Gi1/0/8      1                0            0                Shutdown
Gi1/0/9      1                0            0                Shutdown
Gi1/0/10     1                0            0                Shutdown
Gi1/0/11     1                0            0                Shutdown
Gi1/0/12     1                0            0                Shutdown
Gi1/0/13     1                0            0                Shutdown
Gi1/0/14     1                0            0                Shutdown
Gi1/0/15     1                0            0                Shutdown
Gi1/0/16     1                0            0                Shutdown
Gi1/0/17     1                0            0                Shutdown
Gi1/0/18     1                0            0                Shutdown
Gi1/0/19     1                0            0                Shutdown
Gi1/0/20     1                0            0                Shutdown
Gi1/0/21     1                0            0                Shutdown
Gi1/0/22     1                0            0                Shutdown
Gi1/0/23     1                0            0                Shutdown
Gi1/0/24     1                0            0                Shutdown
Gi1/1/1      1                0            0                Shutdown
Gi1/1/2      1                0            0                Shutdown
Gi1/1/3      1                0            0                Shutdown
Gi1/1/4      1                0            0                Shutdown
Ap1/0/1      1                0            0                Shutdown
-----
Total Addresses in System (excluding one mac per port)    : 1
Max Addresses limit in System (excluding one mac per port) : 4096
```

### 3. show port-security interface g1/0/1 – FL1

FL1 host 1 on the IT VLAN. Port security enabled, and violation mode set to restrict.

```
FL1#show port-security int g1/0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 2
Total MAC Addresses     : 2
Configured MAC Addresses : 0
Sticky MAC Addresses    : 2
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

### 4. show port-security interface g1/0/3 – FL1

FL1 host 2 on the Customer Support VLAN. Port security enabled, and violation mode set to restrict.

```
FL1#show port-security int g1/0/3
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 2
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 24fb.e389.6c43:20
Security Violation Count : 0
```

## 5. show port-security interface g1/0/1 – FL2

FL2 host 1 on the HR VLAN. Port security enabled, and violation mode set to restrict.

```
FL2#show port-security int g1/0/1
Port Security           : Enabled
Port Status             : Secure-down
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 2
Total MAC Addresses     : 2
Configured MAC Addresses : 0
Sticky MAC Addresses    : 2
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

## 6. show port-security interface g1/0/3 – FL2

FL2 host 2 on the Management VLAN. Port security enabled, and violation mode set to restrict.

```
FL2#show port-security int g1/0/3
Port Security           : Enabled
Port Status             : Secure-down
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 2
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 24fb.e389.6c43:50
Security Violation Count : 0
```

## 7. show ip dhcp snooping – FL1

Used access ports have a limit rate of 10 messages per second, unused ports have a limit rate of 1 message per second, and trusted ports have no limit rate.

```
FL1#show ip dhcp snooping
Switch DHCP snooping is enabled
Switch DHCP gleanig is disabled
DHCP snooping is configured on following VLANs:
10,20,30,40,50,60,70,80
DHCP snooping is operational on following VLANs:
10,20,30,40,50,60,70,80
Proxy bridge is configured on following VLANs:
none
Proxy bridge is operational on following VLANs:
none
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is enabled
  circuit-id default format: vlan-mod-port
  remote-id: 4014.825c.3480 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:
```

| Interface             | Trusted | Allow option | Rate limit (pps) |
|-----------------------|---------|--------------|------------------|
| GigabitEthernet1/0/1  | no      | no           | 10               |
| Interface             | Trusted | Allow option | Rate limit (pps) |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/2  | yes     | yes          | unlimited        |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/3  | no      | no           | 10               |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/4  | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/5  | yes     | yes          | unlimited        |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/6  | yes     | yes          | unlimited        |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/7  | yes     | yes          | unlimited        |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/8  | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/9  | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/10 | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/11 | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/12 | no      | no           | 1                |

## 8. show ip dhcp snooping – FL2

Used access ports have a limit rate of 10 messages per second, unused ports have a limit rate of 1 message per second, and trusted ports have no limit rate.

```
FL2#show ip dhcp snooping
Switch DHCP snooping is enabled
Switch DHCP gleaning is disabled
DHCP snooping is configured on following VLANs:
10,20,30,40,50,60,70,80
DHCP snooping is operational on following VLANs:
10,20,30,40,50,60,70,80
Proxy bridge is configured on following VLANs:
none
Proxy bridge is operational on following VLANs:
none
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is enabled
  circuit-id default format: vlan-mod-port
  remote-id: 4014.825a.ac80 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:
```

| Interface             | Trusted | Allow option | Rate limit (pps) |
|-----------------------|---------|--------------|------------------|
| GigabitEthernet1/0/1  | no      | no           | 10               |
| Interface             | Trusted | Allow option | Rate limit (pps) |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/2  | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/3  | no      | no           | 10               |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/4  | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/5  | yes     | yes          | unlimited        |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/6  | yes     | yes          | unlimited        |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/7  | yes     | yes          | unlimited        |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/8  | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/9  | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/10 | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/11 | no      | no           | 1                |
| Custom circuit-ids:   |         |              |                  |
| GigabitEthernet1/0/12 | no      | no           | 1                |

## 9. show ip arp inspection vlan 10-80 – FL1

DAI active on all VLANs on FL1.

```
FL1#show ip arp inspection
Source Mac Validation      : Disabled
Destination Mac Validation : Disabled
IP Address Validation      : Disabled
```

| Vlan | Configuration | Operation | ACL Match | Static ACL |
|------|---------------|-----------|-----------|------------|
| 10   | Enabled       | Active    |           |            |
| 20   | Enabled       | Active    |           |            |
| 30   | Enabled       | Active    |           |            |
| 40   | Enabled       | Active    |           |            |
| 50   | Enabled       | Active    |           |            |
| 60   | Enabled       | Active    |           |            |
| 70   | Enabled       | Active    |           |            |
| 80   | Enabled       | Active    |           |            |

  

| Vlan | ACL Logging | DHCP Logging | Probe Logging |
|------|-------------|--------------|---------------|
| 10   | Deny        | Deny         | Off           |
| 20   | Deny        | Deny         | Off           |
| 30   | Deny        | Deny         | Off           |
| 40   | Deny        | Deny         | Off           |
| 50   | Deny        | Deny         | Off           |

## 10. show ip arp inspection vlan 10-80 – FL2

DAI active on all VLANs on FL2.

```
FL2#show ip arp inspection vlan 10,20,30,40,50,60,70,80
Source Mac Validation      : Disabled
Destination Mac Validation : Disabled
IP Address Validation      : Disabled
```

| Vlan | Configuration | Operation | ACL Match | Static ACL |
|------|---------------|-----------|-----------|------------|
| 10   | Enabled       | Active    |           |            |
| 20   | Enabled       | Active    |           |            |
| 30   | Enabled       | Active    |           |            |
| 40   | Enabled       | Active    |           |            |
| 50   | Enabled       | Active    |           |            |
| 60   | Enabled       | Active    |           |            |
| 70   | Enabled       | Active    |           |            |
| 80   | Enabled       | Active    |           |            |



## 11. show spanning-tree summary – FL1

BPDU Guard is enabled on the FL1 switch.

```
FL1(config)#spanning-tree port
FL1(config)#spanning-tree portfast bpdu
FL1(config)#spanning-tree portfast bpduguard default
FL1(config)#exit
FL1#
*Apr 19 18:37:40.160: %SYS-5-CONFIG_I: Configured from console by console
FL1#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: none
EtherChannel misconfig guard          is enabled
Extended system ID                    is enabled
Portfast Default                      is disabled
PortFast BPDU Guard Default           is enabled
Portfast BPDU Filter Default          is disabled
Loopguard Default                     is disabled
UplinkFast                            is disabled
BackboneFast                           is disabled
Configured Pathcost method used is long
```

| Name     | Blocking | Listening | Learning | Forwarding | STP | Active |
|----------|----------|-----------|----------|------------|-----|--------|
| VLAN0010 | 0        | 0         | 0        | 2          |     | 2      |
| VLAN0020 | 0        | 0         | 0        | 2          |     | 2      |
| VLAN0030 | 0        | 0         | 0        | 2          |     | 2      |
| VLAN0040 | 0        | 0         | 0        | 2          |     | 2      |
| VLAN0050 | 0        | 0         | 0        | 2          |     | 2      |
| VLAN0060 | 0        | 0         | 0        | 2          |     | 2      |
| VLAN0070 | 0        | 0         | 0        | 2          |     | 2      |
| VLAN0080 | 0        | 0         | 0        | 2          |     | 2      |

| Name     | Blocking | Listening | Learning | Forwarding | STP | Active |
|----------|----------|-----------|----------|------------|-----|--------|
| VLAN1000 | 0        | 0         | 0        | 2          |     | 2      |
| 9 vlans  | 0        | 0         | 0        | 18         |     | 18     |

## 12. show spanning-tree summary – FL2

BPDU Guard is enabled on the FL2 switch.

```
FL2#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0010, VLAN0020, VLAN0030, VLAN0040, VLAN0050, VLAN0060
VLAN0070, VLAN0080, VLAN1000
EtherChannel misconfig guard      is enabled
Extended system ID                is enabled
Portfast Default                  is disabled
PortFast BPDU Guard Default       is enabled
Portfast BPDU Filter Default      is disabled
Loopguard Default                 is disabled
UplinkFast                       is disabled
BackboneFast                      is disabled
Configured Pathcost method used is long
```

| Name     | Blocking | Listening | Learning | Forwarding | STP Active |
|----------|----------|-----------|----------|------------|------------|
| VLAN0010 | 0        | 0         | 0        | 1          | 1          |
| VLAN0020 | 0        | 0         | 0        | 1          | 1          |
| VLAN0030 | 0        | 0         | 0        | 1          | 1          |
| VLAN0040 | 0        | 0         | 0        | 1          | 1          |
| VLAN0050 | 0        | 0         | 0        | 1          | 1          |
| VLAN0060 | 0        | 0         | 0        | 1          | 1          |
| VLAN0070 | 0        | 0         | 0        | 1          | 1          |

  

| Name     | Blocking | Listening | Learning | Forwarding | STP Active |
|----------|----------|-----------|----------|------------|------------|
| VLAN0080 | 0        | 0         | 0        | 1          | 1          |
| VLAN1000 | 0        | 0         | 0        | 1          | 1          |

  

| Name    | Blocking | Listening | Learning | Forwarding | STP Active |
|---------|----------|-----------|----------|------------|------------|
| 9 vlans | 0        | 0         | 0        | 9          | 9          |